Aircraft/Airline Low-Risk Potential Analysis

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Overview

- The project aims to identify the low-risk aircraft to support the company entry into exploring expansion into the airline industry.

- This insight will guide the head of the new division in making informed purchasing decisions.

Data & Methods

The NTSB aviation accident database will be used for investigation

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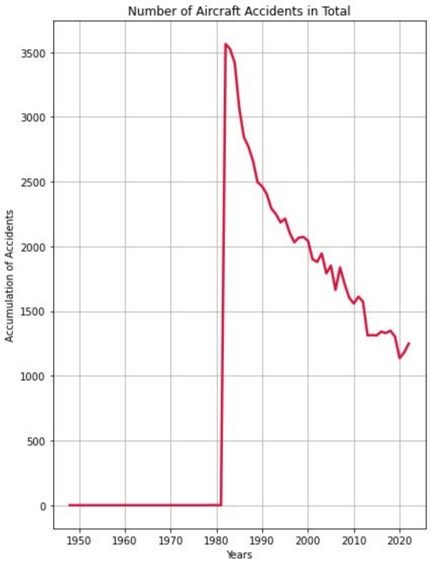
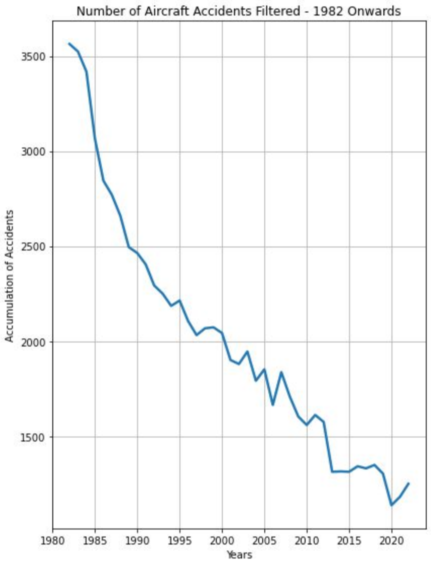
Fours analysis will be explored

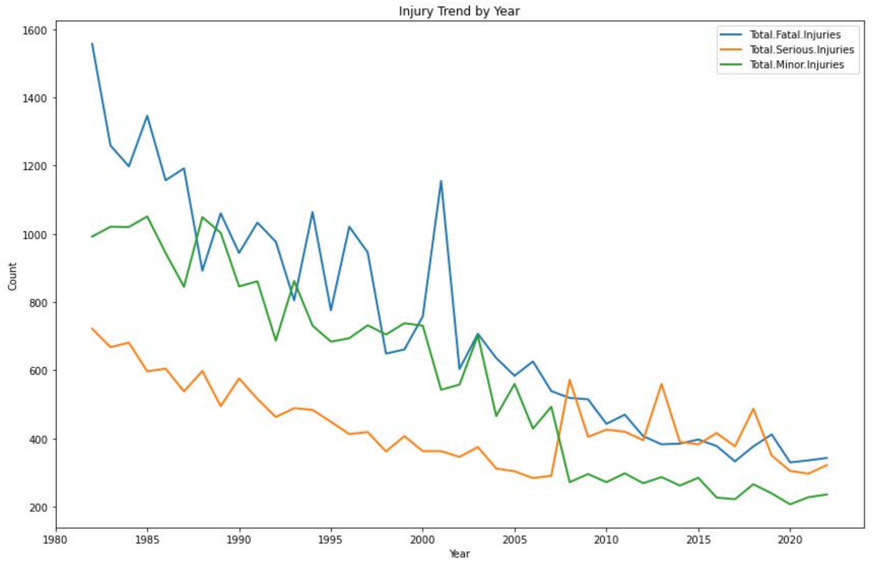
- Data analysis showing the number of aircraft incidents
- Analysis of three data set columns showing Fatal, Serious and Minor injuries for the years 1982 thru 2022
- Analysis showing the Top 10 Manufacture Accidents occurrences
- Analysis of the Top 10 Number of Engine Accidents

Data Analysis - 1

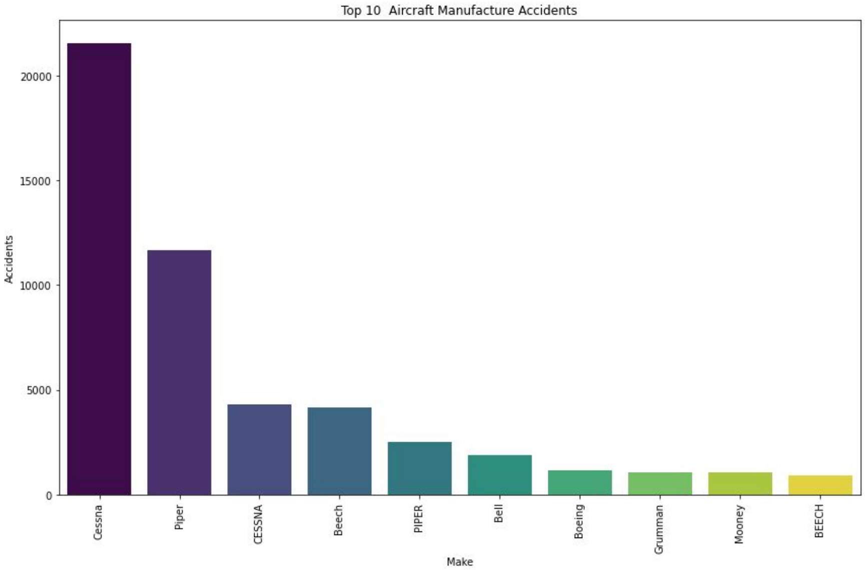




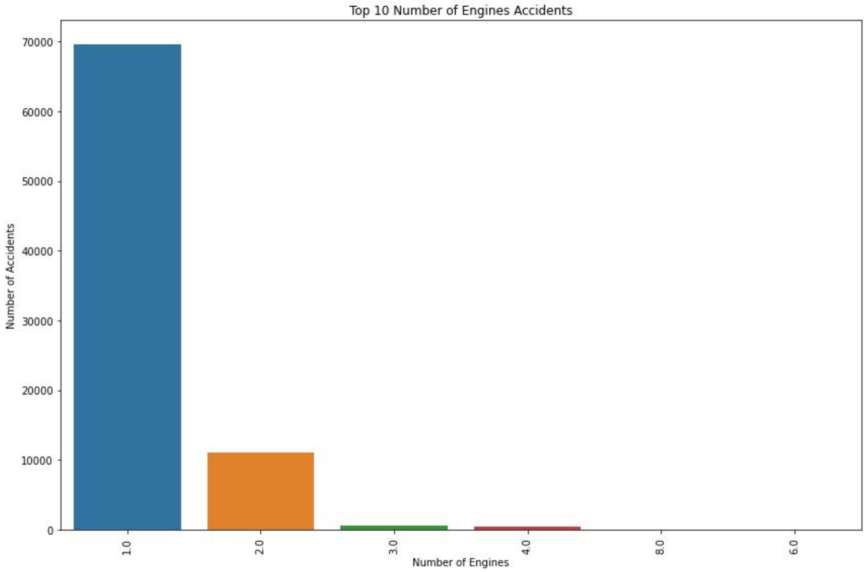
Data Analysis - 2



Data Analysis - 3



Data Analysis - 4



Conclusions

- Analysis is showing that the airline industry shows significant decline in incidences starting from 1985 to present a time, showing improvement in advancement in aviation technology can safety measures.

- Additionally, data shows that single-engine aircraft are more prone to accidents compared to two engine aircrafts. This is likely due to their mechanical limitations and operational vulnerabilities.

- Two engine aircraft provides good balance of power, efficiency and redundancy. This allows for a flight to continue even if one of the engines were to fail.

- These findings highlight the progress in aviation safety and any ongoing risks associated with smaller aircraft.

Limitations

- A clear limitation of this dataset is that the United States is the only country well-represented, while other countries have very few entries. Therefore, any conclusions drawn should primarily focus on aviation trends in the United States.
- Additionally, data shows that single-engine aircraft are more prone to accidents compared to two engine

Recommendations

- To improve insight and decision-making, expanding data collection to include more countries is recommended for a more complete view of global aviation safety.

Next Steps

- Next steps would include collecting more global data, improving accuracy, and analyzing trends to improve aviation safety.

- Next steps would include collaboration with industry stakeholders and regulatory agencies can also reporting standards.

Thanks

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